We are being a little loose with our notation. The vectors of a tensor product space are *new* vectors, built up from the vectors of two smaller spaces. Technically, the same is true for the operators. If we were being pedantic, we would insist on writing the tensor product versions of σ_z and τ_x as $\sigma_z \otimes I$ and $I \otimes \tau_x$, respectively, where I is the identity operator. In fact, we can highlight two important properties of tensor product operators by rewriting the equation

$$\sigma_z|du\rangle = -|du\rangle \qquad (6.9)$$

as

$$\begin{aligned}(\sigma_z \otimes I)\ \ (|d\rangle \otimes |u\rangle) &= (\sigma_z|d\rangle \otimes I|u\rangle) \\ &= (-|d\rangle \otimes |u\rangle). \qquad (6.10)\end{aligned}$$

This notation is cumbersome, and we'll usually stick to the simpler language of Eq. 6.9. However, the language of Eq. 6.10 makes two things clear:

1. A composite operator $\sigma_z \otimes I$ is operating on a composite vector $|d\rangle \otimes |u\rangle$ to produce a new composite vector $-|d\rangle \otimes |u\rangle$.

2. Alice's half (the left half) of the composite operator only affects her half of the composite vector. Likewise, Bob's half of the operator only affects his half of the vector.

We'll have more to say about composite operators in the next section. Furthermore, in Lecture 7, the language of Eq. 6.10 will help us see how to work with tensor products in component form.

Exercise 6.4: Use the matrix forms of σ_z,σ_x, and σ_y and the column vectors for $|u\}$ and $|d\}$ to verify Eqs. 6.6. Then, use Eqs. 6.6 and 6.7 to